

Towards Optimized Cybersickness Prediction for Computationally Constrained Standalone Virtual Reality Devices

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ABSTRACT

Cybersickness, a spectrum of discomforts affecting 60-95% of virtual reality (VR) users, remains a major challenge for comfortable immersive experiences. While recent research using multimodal data and complex machine learning models for cybersickness prediction shows promise, their reliance on external computational resources (cloud servers, GPUs) renders them unsuitable for standalone VR devices (SVRs) due to their limited processing power and potential network lag, which can worsen cybersickness. To bridge this gap, we propose a novel approach that minimizes model complexity and training parameters through hyperparameter tuning, achieving comparable prediction accuracy and error rates with significantly reduced inference time. Our initial findings pave the way for future research on optimized cybersickness prediction models for computationally constrained SVR devices.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality—Cybersickness;

1 INTRODUCTION

Cybersickness, characterized by symptoms such as nausea and dizziness during or after virtual reality (VR) exposure, remains a significant challenge for immersive experiences [4, 7]. Mitigating cybersickness effectively depends on three aspects: real-time severity detection, adaptive mitigation, and addressing post-immersion effects. Traditional methods, such as the Simulator Sickness Questionnaire [4], are limited by their manual nature and lack of real-time capability. Recently, multimodal physiological data combined with complex machine learning models like LSTM, CNN-LSTM, and DeepTCNs have been used for cybersickness prediction [2, 5], but these require substantial external computing resources, making them impractical for standalone VR (SVR) devices due to computational constraints and the risk of network lag aggravating cybersickness [8].

To address this research gap, our research focuses on optimizing cybersickness prediction models for SVR devices, maintaining accuracy while reducing computational demands (i.e., inference time). Motivated by advancements in SVR technology and efficient deep neural model optimization for mobile devices [1, 6], we propose a framework for hyperparameter tuning and computational cost minimization. This approach significantly improves training and inference times, making real-time prediction feasible on resource-limited SVR devices. Our methods, validated using the open-source cybersickness dataset from Islam et al. [2, 3], aim to reduce cloud dependency, enhance data privacy, and offer cost-effective and timely solutions, potentially deployable on various processors, including CPUs, ASICs, FPGAs, and mobile GPUs.

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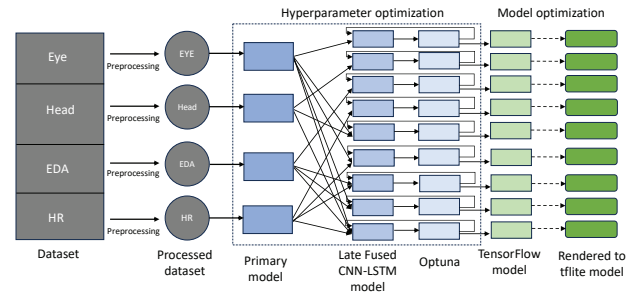


Figure 1: An overview of our proposed approach to optimizing the model for cybersickness prediction by hyperparameter and model's training parameter optimization

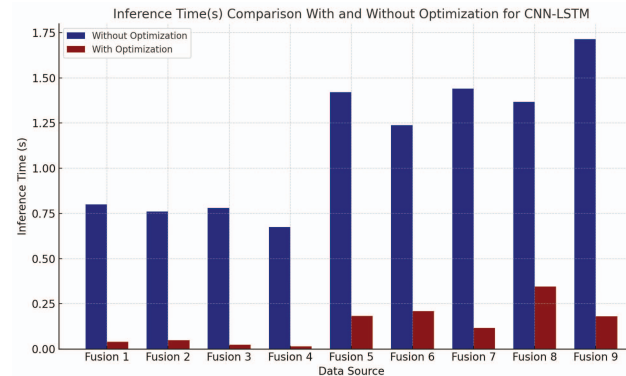


Figure 2: Significant differences in inference time were observed for the optimized models compared to the non-optimized models across various fusion strategies.

2 METHODOLOGY

In this study, we used a dataset provided by Islam et al. [2] that includes eye tracking, head tracking, ectodermal activity (EDA), and heart rate (HR) records from 27 individuals observing various simulations like beaches, rollercoasters, roads, rooms, boats, and shops. The dataset also contained the corresponding fast-motion-scale (FMS) values as the ground truth. The initial step involved processing this dataset by removing outliers (data points more than three standard deviations from the mean) and smoothing the data (moving average). We then proposed a Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) model architecture tailored for the different data types: eye, head, EDA, and HR. For the CNN model, we applied three MaxPooling1D and four convolution1D layers, where the kernel size was 18, and all the layers were bounded in the TimeDistributed Layers. There were seven LSTM layers and one dense layer, followed by the CNN layers. A late-fusion approach was applied to integrate the model parameters from various data modalities and fusion strategies.

Table 1: RMSE and MAE Comparison for CNN-LSTM Model With and Without Optimization

Fusion Strategy	Data Source	RMSE No Opt.	RMSE Opt.	MAE No Opt.	MAE Opt.
Fusion 1	<i>EYE</i>	0.181	0.182	0.153	0.157
Fusion 2	<i>HEAD</i>	0.182	0.182	0.153	0.157
Fusion 3	<i>EDA</i>	0.193	0.192	0.159	0.163
Fusion 4	<i>HR</i>	0.193	0.193	0.159	0.159
Fusion 5	<i>EYE + HEAD</i>	0.123	0.109	0.098	0.088
Fusion 6	<i>EYE + HR</i>	0.229	0.229	0.217	0.217
Fusion 7	<i>EYE + EDA</i>	0.110	0.110	0.089	0.089
Fusion 8	<i>EDA + HR</i>	0.130	0.108	0.108	0.088
Fusion 9	<i>EYE + HEAD + EDA + HR</i>	0.229	0.201	0.217	0.187

We used Optuna to tune the hyperparameters iteratively on the fused models to optimize the models' hyperparameters. In the Optuna, the batch size was 64, 128, and 256; epochs were 100 to 150; dropout was 0.4 to 0.45; and the total trial was five times. The models were then converted to the Tensorflow Lite format, which included further optimization through bit length pruning, enhancing model efficiency. The performance of these models was evaluated based on their inference times (i.e., time required to predict in real-time) and prediction errors. The dataset was divided, with 75% used for training and the remaining 17% and 8% for validation and testing, respectively, across both optimized and non-optimized CNN-LSTM models.

3 RESULTS

We evaluated the models by reporting the root-mean-squared-error (RMSE) and mean-absolute-error (MAE) for predicting FMS values, encompassing both optimized and non-optimized versions across various data fusion strategies (Table 1). Our analysis comprised nine models: four were based on individual data types (EYE, HEAD, EDA, HR), and the other five were composite models resulting from data fusion (Table 1). The comparative assessment revealed no significant disparity in RMSE and MAE values between optimized and non-optimized models, indicating that the optimization process effectively maintains the models' predictive performance.

To evaluate the impact of optimization on inference time, we compared the performance of both non-optimized and optimized models, with results illustrated in Figure 2. Notably, optimized models significantly reduced inference time compared to their non-optimized counterparts. For the optimized eye data (Fusion 1), inference time was decreased to 0.04s from 0.8s, marking a 95% improvement. Similarly, for the fully fused model (Fusion 9), the optimized version required only 0.181s for inference, an improvement from 1.715s in the non-optimized model (i.e., 89.4% improvement). This trend of significant efficiency gains with optimization was evident across other models as well: for the head data (Fusion 2), the improvement was from 0.761s to 0.049s (93.6% improvement), for EDA data (fusion 3) improvement was from 0.781s to 0.024s (96.9% improvement), for HR data (fusion 4) improvement was from 0.675s to 0.014s (97.9% improvement), for eye and head data (fusion 5) improvement was from 1.42s to 0.183s (87.1% improvement), for eye and HR data (fusion 6) improvement was from 1.24s to 0.413s (66.7% improvement), for eye and eda data (fusion 7) improvement was from 1.441s to 0.117s (91.9% improvement), for eda and hr data (fusion 8) improvement was from 1.367s to 0.345s (74.8% improvement), and for the combination of eye, head, EDA, and HR which is fusion ten improvement was from 1.715s to 0.181s (89.4% improvement). These results highlight the substantial enhancements in model performance due to optimization.

4 CONCLUSION & FUTURE WORK

In summary, our research involved processing individual datasets to create separate models, which were fused and optimized for improved inference times. We conducted a comparative analysis of RMSE, MAE, and inference time before and after optimization, observing no substantial changes in error metrics but significant improvements in inference speed. However, our study faced limitations in testing the models' real-time inference capabilities. Future work will focus on deploying these models on stand-alone VR devices to assess their on-time inference time and latency, providing a more comprehensive understanding of their performance in real-world scenarios.

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